

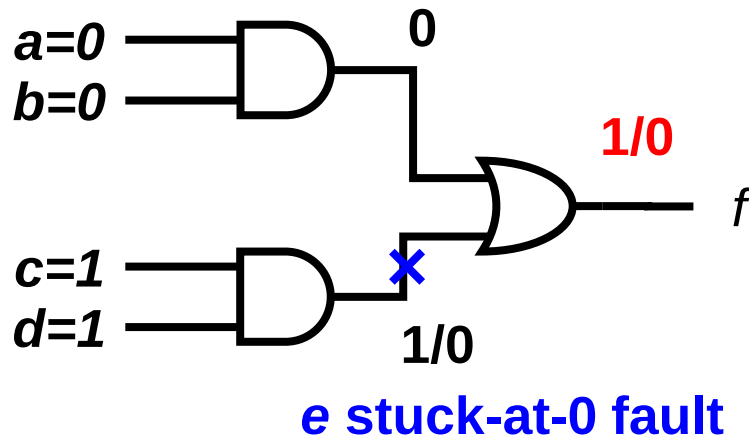
Fault Modeling

- Introduction
- Fault Models
- **Fault Detection**
- Fault Coverage
- Conclusion



Fault Detection

- A test pattern **detects** a fault if
 - ♦ output of faulty circuit $\neq$ output of good circuit



$$f = ab + cd$$

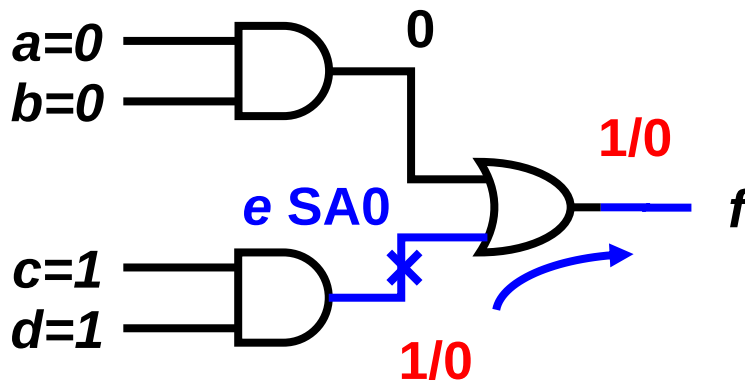
test pattern $(a,b,c,d) = (0,0,1,1)$

detects e stuck-at-0 fault

notation 1/0:
good value=1; faulty value =0

Activation & Propagation

- A fault is detected if two conditions are satisfied
 - ◆ (1) **Fault activation**:
 - * different value at **fault site** (opposite to faulty value)
 - ◆ (2) **Fault effect propagation**: (aka. *error propagation*)
 - * Propagate *fault effect* to any primary output
 - * Signal x is *sensitized* if output changes when x changes
 - * A path of sensitized signals is called *sensitized path*



Fault activation: $c=d=1$
Fault Effect Propagation: $a=b=0$
e is sensitized to output
e-f is sensitized path

Detection Requires (1) Activation & (2) Propagation

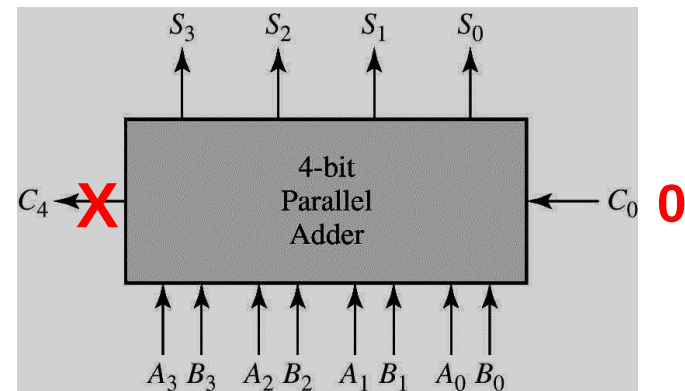
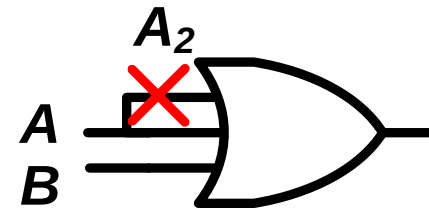
Fault Classes

- **Unstable faults*** = faults that cannot be detected by any test pattern
- **Testable faults** = faults that cannot be proven unstable
 - ♦ **Detected faults**
 - ♦ **Undetected faults**
 - ♦ Oscillatory faults (see fault simulation)
 - ♦ Potentially detected faults (see fault simulation)
 - ♦ ...

* **unstable** faults is aka. **undetectable** faults
but very confusing so not used in lecture

Untestable Faults

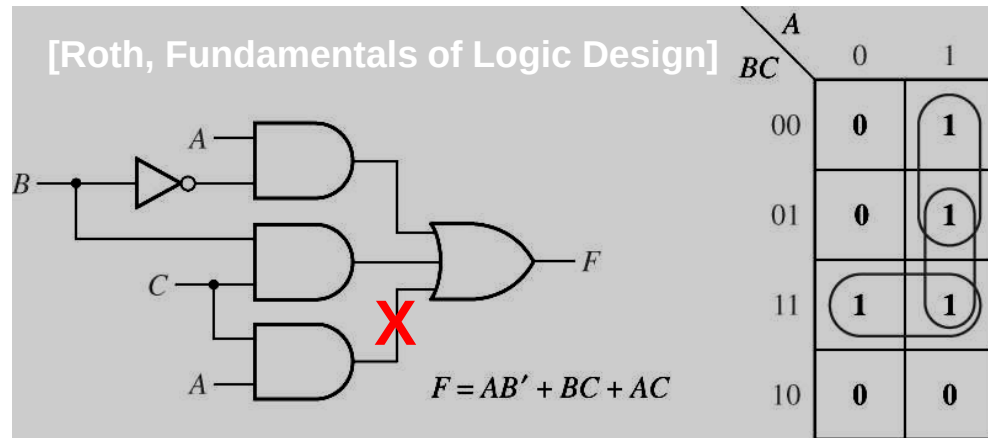
- Definition
 - ♦ Faults that cannot be detected by any test pattern (proven)
 - ♦ Aka *redundant faults*
- Proving untestable fault is *NP-complete*
 - ♦ same as *satisfiability (SAT)*
- Examples of untestable faults
 - ♦ 1. Redundant circuitry
 - * A_2 stuck-at zero fault is untestable
 - ♦ 2. Unused output or tied input
 - * C_4 is not used
 - SA0, SA1 untestable
 - * C_0 is tied to zero
 - SA0 untestable



Untestable Faults (2)

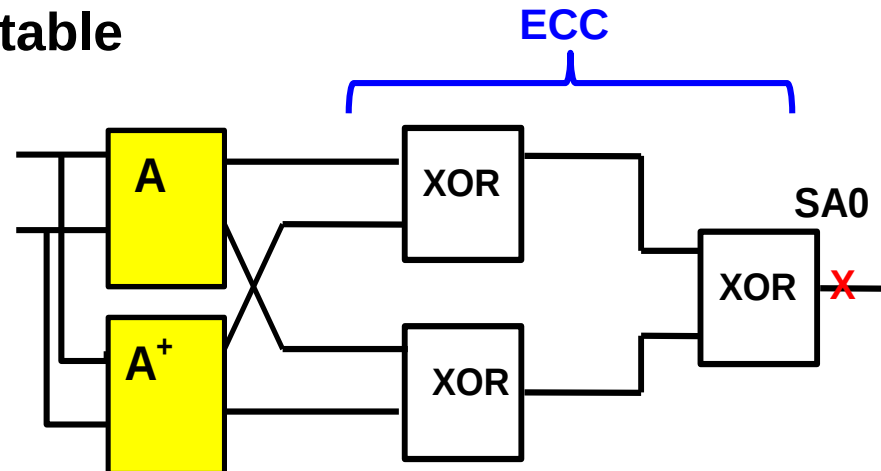
- 3. Hazard control circuitry

- ◆ $AB' + BC + \underline{AC} = AB' + BC$
- ◆ AC SA0 is untestable
- ◆ Used for hazard prevention



- 4. Error control circuitry

- ◆ A and A⁺ always produce same output
- ◆ ECC output SA0 is untestable



- 5. ATPG constraints

- ◆ e.g. set PI=1

Undetected Faults

- Definition
 - ◆ Faults that are **not detected by the given test set**
- Due to ATPG runtime limitation
 - ◆ Cannot prove it untestable
 - ◆ Cannot find test pattern, either
- NOTE: do not confuse *untestable faults* with *undetected faults*
 - ◆ Former: no test pattern exists (proven)
 - ◆ Latter: no test pattern so far (may exist but not sure)

Untestable Faults $\neq$ Undetected Faults

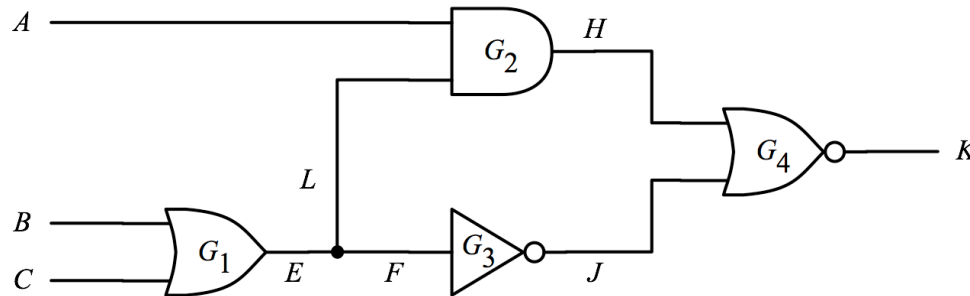
Quiz

Q1: Apply two patterns {000, 001}. Which fault (s) are undetected?

A:

Q2: Now consider all patterns. Which fault(s) are untestable?

A:

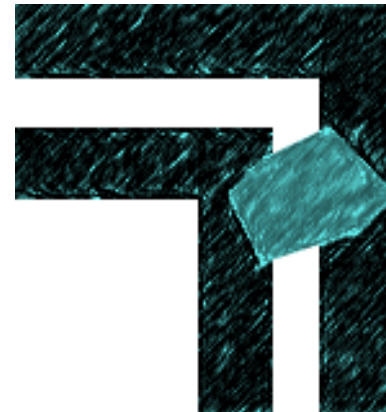


Input			Output						
A	B	C	good	<i>E/0</i>	<i>F/0</i>	<i>L/0</i>	<i>E/1</i>	<i>F/1</i>	<i>L/1</i>
0	0	0	0	0	0	0	1	1	0
0	0	1	1	0	0	1	1	1	1
0	1	0	1	0	0	1	1	1	1
0	1	1	1	0	0	1	1	1	1
1	0	0	0	0	0	0	0	1	0
1	0	1	0	0	0	1	0	0	0
1	1	0	0	0	0	1	0	0	0
1	1	1	0	0	0	1	0	0	0

Consider only
six faults on *EFL*

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More Metrics

- **Fault coverage (FC):** measure quality of test set

$$\frac{\text{number of } \textit{detected} \text{ faults}}{\text{number of } \textit{total} \text{ faults}} \times 100\%$$

- **Test coverage :** measure quality of test set (testable faults only)

$$\frac{\text{number of } \textit{detected} \text{ faults}}{\text{number of } \textit{testable} \text{ faults}} \times 100\%$$

- **ATPG effectiveness :** measure quality of ATPG algorithm

$$\frac{\text{number of } \textit{detected} + \textit{untestable} \text{ faults}}{\text{number of } \textit{total} \text{ faults}} \times 100\%$$

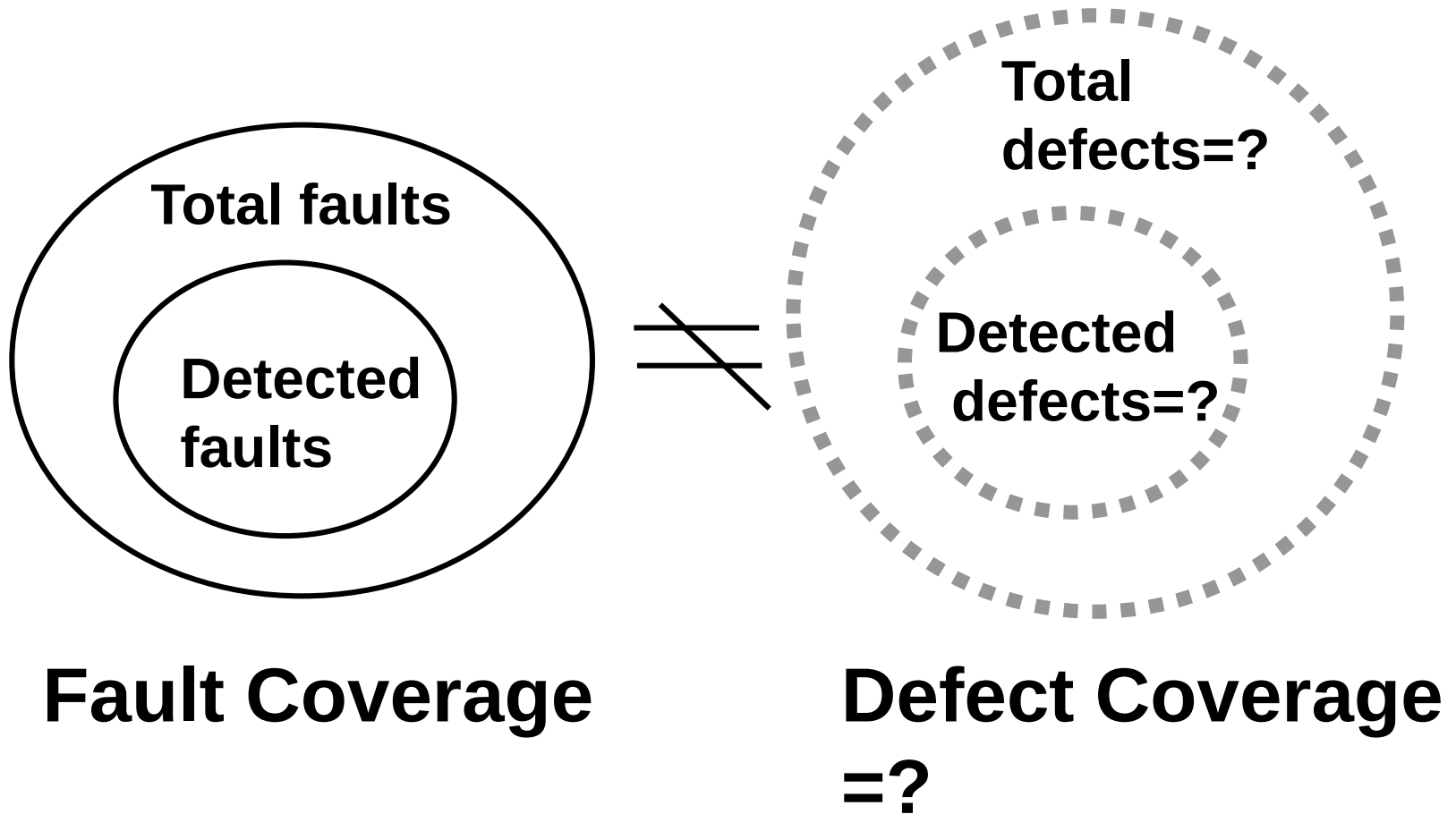
*NOTE: Each ATPG has its own definition. Details see ATPG manual.

Quiz

Item	Number
Total Faults	1,234
Detected faults	1,000
Untestable faults	230
Undetected faults	4
Fault Coverage	
Test Coverage	
ATPG effectiveness	

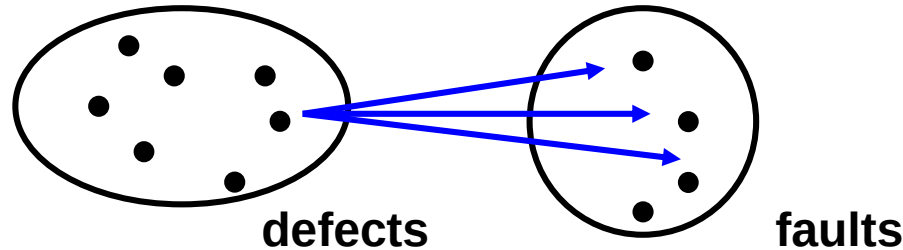
Revisit: $DL = 1 - Y^{(1-FC)}$

- In practice, does 100% FC mean 0 DPM?
 - ♦ **NO!** Fault coverage does **NOT** represent defect coverage



Defect $\rightarrow$ Faults

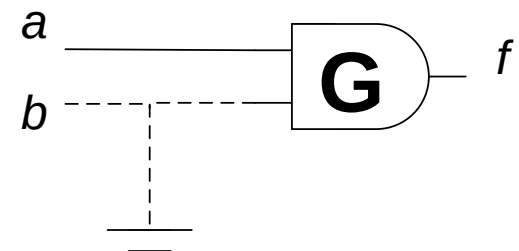
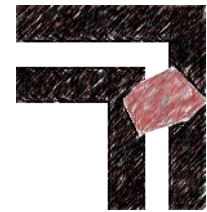
- Sometimes, one defect can be modeled by more than one fault



- Example:

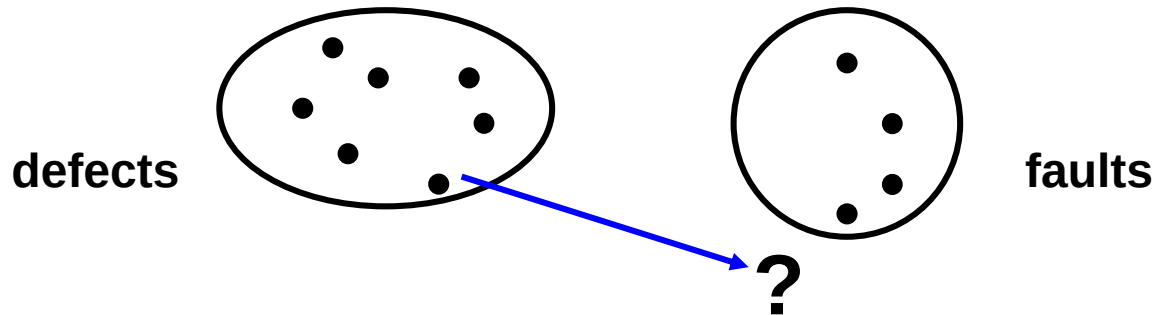
- ♦ unwanted wire between b and ground

- * b stuck-at zero SSF
- * f stuck-at zero SSF
- * b slow-to-rise transition fault
- * Gate delay fault in G
- * ...



Defect → Faults (2)

- Sometimes, one defect cannot be well modeled by any fault



- Examples:
 - ♦ Slow process cannot be well modeled by SSF
 - * (maybe path delay fault)
 - ♦ Reliability defects cannot be well modeled by SSF
 - ♦ Fault masking
 - ♦ ...

Experimental Results

- 0.7 μ m CMOS, **Murphy experiment, Stanford Univ.** [McCluskey 00]
 - ◆ Total population 5.5K chips tested
 - * 116 defective chips
 - * **Only 1/3** of defects behaves like SSF
 - * **2~6 chips escaped 100% SSF test sets**
- ***N-detect*** SSF test patterns
 - ◆ Detect each SSF at least N times by different patterns
 - ◆ **No chip escaped $N=3$**
 - ◆ Why? Accidental detection of unmolded defects

100% FC $\neq$ 0 DPM

Diversified Test Patterns Are Good

Fault Modeling

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Concluding Remarks

- Fault model is very important for test automation
 - ◆ Automatic test pattern generation
 - ◆ Quantify quality of test patterns
- Although many fault models, only a few used in practice
 - ◆ **Single stuck-at faults** is applied for sure
 - ◆ **Transition delay faults** may be applied
 - ◆ Other fault models adopted according to product needs
- **NOTICE! Fault model can be deceiving**
 - ◆ Fault models do not always match behavior of defects
 - * **Only 1/3** of defects behaves like SSF
 - ◆ 100% fault coverage DOES NOT guarantee zero DPM

Isn't 99.9% Good Enough?

- Look at the consequences of “almost, but not quite” perfect.
- If 99.9% is good enough then:
 - ◆ 2 million documents will be lost by IRS this year
 - ◆ 12 babies will be given to the wrong parents today
 - ◆ 2 plane landings daily at LAX airport will be unsafe
 - ◆ 18K pieces of mail will be mishandled in the next hour
 - ◆ ...

**Because There are Many Faults
Small Δ in FC Makes Large Difference**

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